

Nonparametric Statistics

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Contents

1	Chapter 1: Basic Probability and Asymptotic Tools	3
1.1	Probability, distributions, and conditional quantities	3
1.2	Useful inequalities	4
1.3	Some distribution facts	4
1.4	Modes of stochastic convergence	5
1.5	Main limit theorems	5
1.6	O , o , $O_{\mathbb{P}}$, and $o_{\mathbb{P}}$	6
1.7	Analytical tools	7
2	Chapter 2: Univariate Kernel Density Estimation	7
2.1	Histogram	8
2.2	Moving histogram	8
2.3	Kernel density estimator	9
2.4	Assumptions for the asymptotic analysis	10
2.5	Exact expectation and variance	10
2.6	Asymptotic bias, variance, and MSE	10
2.7	Asymptotic normality	11
2.8	Integrated error criteria	11
2.9	Bandwidth selection	12
2.9.1	Normal scale and rule of thumb	12
2.9.2	Direct plug-in selector	12
2.9.3	Least-squares cross-validation	13
2.9.4	Biased cross-validation	13
2.9.5	Comparison	13
2.10	Boundary correction by transformation	14
2.11	Sampling from a KDE	14
3	Chapter 4: Univariate Kernel Regression	14
3.1	Nadaraya–Watson estimator	14
3.2	Local polynomial estimator	15

3.3	Regression model and assumptions	16
3.4	Conditional bias and variance	16
3.5	Asymptotic normality	17
3.6	Bandwidth selection for local polynomial regression	17
3.6.1	Rule-of-thumb selector	17
3.6.2	Cross-validation	18
3.7	Naive regression estimator and regressogram	18
3.8	Behavior in low-density regions	19
4	Appendix A: Pointwise Confidence Intervals for a Density	19
4.1	Why a direct interval for $f(x)$ is difficult	19
4.2	Normal convolution example	19
4.3	Asymptotic pointwise interval	19

1 Chapter 1: Basic Probability and Asymptotic Tools

1.1 Probability, distributions, and conditional quantities

A probability space is $(\Omega, \mathcal{A}, \mathbb{P})$. A random variable is a measurable map $X : \Omega \rightarrow \mathbb{R}$. The cumulative distribution function and empirical distribution function are

$$F(x) = \mathbb{P}(X \leq x), \quad F_n(x) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}_{\{X_i \leq x\}}.$$

If F is differentiable, its density is $f = F'$ and

$$F(x) = \int_{-\infty}^x f(t) dt.$$

For a measurable function g ,

$$\mathbb{E}[g(X)] = \int g(x) dF(x) = \begin{cases} \int g(x) f(x) dx, & X \text{ continuous,} \\ \sum_x g(x) \mathbb{P}(X = x), & X \text{ discrete.} \end{cases}$$

Also,

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}X)^2] = \mathbb{E}[X^2] - (\mathbb{E}X)^2.$$

For a continuous random vector $\mathbf{X} = (X_1, \dots, X_p)^\top$ with joint density f ,

$$f_{X_i}(x_i) = \int_{\mathbb{R}^{p-1}} f(\mathbf{x}) d\mathbf{x}_{-i}, \quad f_{X_1|\mathbf{X}_{-1}=\mathbf{x}_{-1}}(x_1) = \frac{f(\mathbf{x})}{f_{\mathbf{X}_{-1}}(\mathbf{x}_{-1})}.$$

The conditional mean and variance are

$$\mathbb{E}[Y | X] = \int y dF_{Y|X}(y | X), \quad \text{Var}(Y | X) = \mathbb{E}[Y^2 | X] - \mathbb{E}[Y | X]^2.$$

Laws of total expectation and variance. If the required moments are finite,

$$\mathbb{E}[Y] = \mathbb{E}\{\mathbb{E}[Y | X]\},$$

$$\text{Var}(Y) = \mathbb{E}\{\text{Var}(Y | X)\} + \text{Var}\{\mathbb{E}[Y | X]\}.$$

For two random variables,

$$\text{Cov}(X_1, X_2) = \mathbb{E}[X_1 X_2] - \mathbb{E}[X_1] \mathbb{E}[X_2], \quad \text{Cor}(X_1, X_2) = \frac{\text{Cov}(X_1, X_2)}{\sqrt{\text{Var}(X_1) \text{Var}(X_2)}}.$$

For a random vector $\mathbf{X}$,

$$\text{Var}(\mathbf{X}) = \mathbb{E}[(\mathbf{X} - \mathbb{E}\mathbf{X})(\mathbf{X} - \mathbb{E}\mathbf{X})^\top].$$

If A is a fixed matrix and $\mathbf{b}$ is fixed, then

$$\mathbb{E}[A\mathbf{X} + \mathbf{b}] = A\mathbb{E}[\mathbf{X}] + \mathbf{b}, \quad \text{Var}(A\mathbf{X} + \mathbf{b}) = A \text{Var}(\mathbf{X}) A^\top.$$

1.2 Useful inequalities

Markov. For $X \geq 0$ and $t > 0$,

$$\mathbb{P}(X \geq t) \leq \frac{\mathbb{E}[X]}{t}.$$

Chebyshev. If $\mathbb{E}[X] = \mu$ and $\text{Var}(X) = \sigma^2 < \infty$, then

$$\mathbb{P}(|X - \mu| \geq t) \leq \frac{\sigma^2}{t^2}, \quad t > 0.$$

Equivalently,

$$\mathbb{P}(\mu - t\sigma < X < \mu + t\sigma) \geq 1 - \frac{1}{t^2}.$$

Cauchy–Schwarz. If $\mathbb{E}[X^2], \mathbb{E}[Y^2] < \infty$, then

$$|\mathbb{E}[XY]| \leq \sqrt{\mathbb{E}[X^2]\mathbb{E}[Y^2]}.$$

Jensen. If g is convex, then

$$g(\mathbb{E}[X]) \leq \mathbb{E}[g(X)].$$

In particular, for $r \geq 1$,

$$|\mathbb{E}[X]|^r \leq \mathbb{E}[|X|^r].$$

If $0 < r \leq s$, then $\mathbb{E}|X|^s < \infty$ implies $\mathbb{E}|X|^r < \infty$.

1.3 Some distribution facts

For $X \sim N(\mu, \sigma^2)$,

$$\phi_\sigma(x - \mu) = \frac{1}{\sigma} \phi\left(\frac{x - \mu}{\sigma}\right), \quad \mathbb{P}(X \in \mu \pm z_{\alpha/2}\sigma) = 1 - \alpha,$$

where $z_{\alpha/2} = \Phi^{-1}(1 - \alpha/2)$. Also,

$$\mathbb{E}[X] = \mu, \quad \mathbb{E}[X^2] = \mu^2 + \sigma^2, \quad \mathbb{E}[X^3] = \mu^3 + 3\mu\sigma^2, \quad \mathbb{E}[X^4] = \mu^4 + 6\mu^2\sigma^2 + 3\sigma^4.$$

For $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \Sigma)$,

$$\phi_\Sigma(\mathbf{x} - \boldsymbol{\mu}) = (2\pi)^{-p/2} |\Sigma|^{-1/2} \exp\left\{-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})\right\},$$

$$A\mathbf{X} + \mathbf{b} \sim N_q(A\boldsymbol{\mu} + \mathbf{b}, A\Sigma A^\top).$$

A compact list of other facts used later is

Distribution	Main facts
$X \sim \text{LN}(\mu, \sigma^2)$	$X \stackrel{d}{=} \exp(Z)$ for $Z \sim N(\mu, \sigma^2)$; $\mathbb{E}[X] = e^{\mu + \sigma^2/2}$ and $\text{Var}(X) = (e^{\sigma^2} - 1)e^{2\mu + \sigma^2}$.
$X \sim \text{Exp}(\lambda)$	$f(x) = \lambda e^{-\lambda x}$ for $x > 0$.
$X \sim \Gamma(a, p)$	Rate a , shape p ; $\mathbb{E}[X] = p/a$ and $\text{Var}(X) = p/a^2$.
$X \sim \text{B}(n, p)$	$\mathbb{E}[X] = np$ and $\text{Var}(X) = np(1 - p)$; $\text{B}(1, p) = \text{Ber}(p)$.
$X \sim \text{Pois}(\lambda)$	$\mathbb{E}[X] = \text{Var}(X) = \lambda$.

1.4 Modes of stochastic convergence

Let X_n and X be random variables.

Convergence in distribution.

$$X_n \xrightarrow{d} X \iff F_n(x) \rightarrow F(x) \text{ at every continuity point of } F.$$

Convergence in probability.

$$X_n \xrightarrow{\mathbb{P}} X \iff \mathbb{P}(|X_n - X| > \varepsilon) \rightarrow 0, \quad \forall \varepsilon > 0.$$

Almost sure convergence.

$$X_n \xrightarrow{\text{a.s.}} X \iff \mathbb{P}\{\omega : X_n(\omega) \rightarrow X(\omega)\} = 1.$$

Convergence in r -mean. For $r \geq 1$,

$$X_n \xrightarrow{r} X \iff \mathbb{E}|X_n - X|^r \rightarrow 0.$$

The basic implication diagram is

$$X_n \xrightarrow{r} X \implies X_n \xrightarrow{\mathbb{P}} X \iff X_n \xrightarrow{\text{a.s.}} X, \quad X_n \xrightarrow{\mathbb{P}} X \implies X_n \xrightarrow{d} X.$$

If the limit is a constant c , then

$$X_n \xrightarrow{d} c \implies X_n \xrightarrow{\mathbb{P}} c.$$

An estimator $\hat{\theta}_n$ is mean-square consistent if

$$\text{MSE}(\hat{\theta}_n) = \mathbb{E}[(\hat{\theta}_n - \theta)^2] = \text{Bias}(\hat{\theta}_n)^2 + \text{Var}(\hat{\theta}_n) \rightarrow 0.$$

1.5 Main limit theorems

Law of large numbers. If $X_1, X_2, \dots$ are iid and $\mathbb{E}[X_i] = \mu$, then

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{\mathbb{P}} \mu, \quad \bar{X}_n \xrightarrow{\text{a.s.}} \mu.$$

Central limit theorem. If the variables are iid with mean μ and variance $\sigma^2 < \infty$, then

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} N(0, 1).$$

Lyapunov CLT. For independent variables with means μ_i , variances σ_i^2 , and $s_n^2 = \sum_{i=1}^n \sigma_i^2$, if for some $\delta > 0$,

$$\frac{1}{s_n^{2+\delta}} \sum_{i=1}^n \mathbb{E}|X_i - \mu_i|^{2+\delta} \rightarrow 0,$$

then

$$\frac{1}{s_n} \sum_{i=1}^n (X_i - \mu_i) \xrightarrow{d} N(0, 1).$$

Cramér–Wold. For random vectors $\mathbf{X}_n, \mathbf{X} \in \mathbb{R}^p$,

$$\mathbf{X}_n \xrightarrow{d} \mathbf{X} \iff \mathbf{c}^\top \mathbf{X}_n \xrightarrow{d} \mathbf{c}^\top \mathbf{X} \text{ for every } \mathbf{c} \in \mathbb{R}^p.$$

Continuous mapping. If g is continuous, then

$$X_n \xrightarrow{d, \mathbb{P}, \text{a.s.}} X \implies g(X_n) \xrightarrow{d, \mathbb{P}, \text{a.s.}} g(X).$$

Slutsky. If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{\mathbb{P}} c$, then

$$X_n + Y_n \xrightarrow{d} X + c, \quad X_n Y_n \xrightarrow{d} cX, \quad \frac{X_n}{Y_n} \xrightarrow{d} \frac{X}{c} \quad (c \neq 0).$$

Delta method. If

$$\sqrt{n}(X_n - \mu) \xrightarrow{d} N(0, \sigma^2)$$

and g is differentiable at μ with $g'(\mu) \neq 0$, then

$$\sqrt{n}\{g(X_n) - g(\mu)\} \xrightarrow{d} N(0, g'(\mu)^2 \sigma^2).$$

1.6 O , o , $O_{\mathbb{P}}$, and $o_{\mathbb{P}}$

For positive deterministic sequences a_n, b_n ,

$$a_n = O(b_n) \iff \limsup_{n \rightarrow \infty} \frac{a_n}{b_n} < \infty,$$

$$a_n = o(b_n) \iff \frac{a_n}{b_n} \rightarrow 0.$$

Thus,

$$a_n = O(b_n) \iff \frac{a_n}{b_n} = O(1), \quad a_n = o(b_n) \iff \frac{a_n}{b_n} = o(1).$$

For a random sequence X_n and a positive deterministic sequence a_n ,

$$X_n = o_{\mathbb{P}}(a_n) \iff \frac{|X_n|}{a_n} \xrightarrow{\mathbb{P}} 0,$$

$$X_n = O_{\mathbb{P}}(a_n) \iff \frac{X_n}{a_n} = O_{\mathbb{P}}(1),$$

where $O_{\mathbb{P}}(1)$ means bounded in probability.

The most frequently used rules are

$$\begin{aligned} o_{\mathbb{P}}(a_n) &\subset O_{\mathbb{P}}(a_n), \\ O_{\mathbb{P}}(a_n) + O_{\mathbb{P}}(b_n) &= O_{\mathbb{P}}(a_n + b_n), \\ o_{\mathbb{P}}(a_n) + o_{\mathbb{P}}(b_n) &= o_{\mathbb{P}}(a_n + b_n), \\ O_{\mathbb{P}}(a_n)O_{\mathbb{P}}(b_n) &= O_{\mathbb{P}}(a_n b_n), \\ o_{\mathbb{P}}(a_n)O_{\mathbb{P}}(b_n) &= o_{\mathbb{P}}(a_n b_n), \\ o_{\mathbb{P}}(1)O_{\mathbb{P}}(a_n) &= o_{\mathbb{P}}(a_n), \\ (1 + o_{\mathbb{P}}(1))^{-1} &= O_{\mathbb{P}}(1). \end{aligned}$$

A useful inequality is

$$|a + b|^p \leq C_p(|a|^p + |b|^p), \quad C_p = \begin{cases} 1, & 0 < p \leq 1, \\ 2^{p-1}, & p > 1. \end{cases}$$

Chebyshev's inequality gives the very useful order relation

$$X_n - \mathbb{E}[X_n] = O_{\mathbb{P}}\left(\sqrt{\text{Var}(X_n)}\right).$$

Hence, a rough but convenient decomposition is

$$X_n = \mathbb{E}[X_n] + O_{\mathbb{P}}\left(\sqrt{\text{Var}(X_n)}\right).$$

1.7 Analytical tools

Mean value theorem. For a differentiable f on (a, b) ,

$$f(b) - f(a) = f'(c)(b - a) \quad \text{for some } c \in (a, b).$$

Integral mean value theorem. For continuous f ,

$$\int_a^b f(x) dx = f(c)(b - a) \quad \text{for some } c \in (a, b).$$

Taylor expansion. If f has p continuous derivatives near x , then

$$f(x + h) = \sum_{j=0}^p \frac{f^{(j)}(x)}{j!} h^j + o(h^p).$$

If $f^{(p+1)}$ is bounded locally, the remainder can be controlled uniformly.

Dominated convergence theorem. If $f_n(x) \rightarrow f(x)$ and $|f_n(x)| \leq g(x)$ with $\int |g| < \infty$, then

$$\lim_{n \rightarrow \infty} \int f_n(x) dx = \int f(x) dx.$$

Parametric versus nonparametric inference. Parametric methods gain efficiency from a finite-dimensional model. Nonparametric methods use weaker assumptions and retain validity over a wider class of distributions. The usual trade-off is

$$\begin{array}{c} \text{stronger model assumptions} \\ \Updownarrow \\ \text{more efficiency, but less protection against misspecification} \end{array}$$

2 Chapter 2: Univariate Kernel Density Estimation

Let $X_1, \dots, X_n$ be iid from a continuous density f with cdf F .

2.1 Histogram

Since

$$f(x_0) = F'(x_0) = \lim_{h \downarrow 0} \frac{F(x_0 + h) - F(x_0)}{h} = \lim_{h \downarrow 0} \frac{\mathbb{P}(x_0 < X < x_0 + h)}{h},$$

a histogram estimates local probability per unit length.

Choose an origin t_0 and a bin width $h > 0$. Define

$$t_k = t_0 + hk, \quad B_k = [t_k, t_{k+1}), \quad k \in \mathbb{Z}.$$

For $x \in B_k$,

$$\hat{f}_H(x; t_0, h) = \frac{1}{nh} \sum_{i=1}^n \mathbf{1}_{\{X_i \in B_k\}} = \frac{v_k}{nh},$$

where $v_k = \#\{i : X_i \in B_k\}$.

Let

$$p_k = \mathbb{P}(X \in B_k) = \int_{B_k} f(t) dt.$$

Then

$$v_k \sim B(n, p_k).$$

If f is continuous, the integral mean value theorem gives

$$p_k = hf(\xi_{k,h}) \quad \text{for some } \xi_{k,h} \in (t_k, t_{k+1}).$$

Therefore,

$$\begin{aligned} \mathbb{E}[\hat{f}_H(x; t_0, h)] &= f(\xi_{k,h}), \\ \text{Var}\{\hat{f}_H(x; t_0, h)\} &= \frac{f(\xi_{k,h})\{1 - hf(\xi_{k,h})\}}{nh}. \end{aligned}$$

If $h \rightarrow 0$, then $\xi_{k,h} \rightarrow x$ and the bias disappears. To make the variance vanish, one also needs

$$nh \rightarrow \infty.$$

The histogram depends on both t_0 and h .

2.2 Moving histogram

Using the symmetric derivative,

$$f(x) = \lim_{h \downarrow 0} \frac{F(x+h) - F(x-h)}{2h} = \lim_{h \downarrow 0} \frac{\mathbb{P}(x-h < X < x+h)}{2h}.$$

The moving histogram, also called the naive density estimator, is

$$\hat{f}_N(x; h) = \frac{1}{2nh} \sum_{i=1}^n \mathbf{1}_{\{x-h < X_i < x+h\}}.$$

Let

$$p_{x,h} = F(x+h) - F(x-h).$$

Then

$$\sum_{i=1}^n \mathbf{1}_{\{x-h < X_i < x+h\}} \sim B(n, p_{x,h}),$$

so

$$\mathbb{E}[\widehat{f}_N(x; h)] = \frac{F(x+h) - F(x-h)}{2h},$$

$$\text{Var}\{\widehat{f}_N(x; h)\} = \frac{F(x+h) - F(x-h)}{4nh^2} - \frac{\{F(x+h) - F(x-h)\}^2}{4nh^2}.$$

For small h ,

$$\text{Var}\{\widehat{f}_N(x; h)\} \approx \frac{f(x)}{2nh} - \frac{f(x)^2}{n}.$$

Thus the basic consistency conditions are again

$$h \rightarrow 0, \quad nh \rightarrow \infty.$$

The moving histogram is the kernel estimator obtained from the rectangular kernel

$$K(z) = \frac{1}{2} \mathbf{1}_{\{|z| < 1\}}.$$

It uses local data, but all observations inside $(x-h, x+h)$ receive the same weight.

2.3 Kernel density estimator

Define the scaled kernel

$$K_h(z) = \frac{1}{h} K\left(\frac{z}{h}\right).$$

The kernel density estimator is

$$\widehat{f}(x; h) = \frac{1}{n} \sum_{i=1}^n K_h(x - X_i) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - X_i}{h}\right).$$

Usually K is a symmetric density with its mode at zero.

Common kernels are

$$\begin{aligned} \text{Normal:} \quad & K(z) = \phi(z), \\ \text{Epanechnikov:} \quad & K(z) = \frac{3}{4}(1 - z^2) \mathbf{1}_{\{|z| < 1\}}, \\ \text{Rectangular:} \quad & K(z) = \frac{1}{2} \mathbf{1}_{\{|z| < 1\}}. \end{aligned}$$

For the normal kernel,

$$K_h(x - X_i) = \phi_h(x - X_i),$$

so $\widehat{f}$ is an equally weighted mixture of $N(X_i, h^2)$ densities.

Useful notation:

$$\mu_2(K) = \int z^2 K(z) dz, \quad R(g) = \int g(x)^2 dx,$$

$$(f * g)(x) = \int f(x - y)g(y) dy.$$

Once the kernel is symmetric and regular enough, the choice of h matters much more than the exact shape of K .

2.4 Assumptions for the asymptotic analysis

- A1.** f is square integrable, twice continuously differentiable, and f'' is square integrable.
A2. K is a symmetric and bounded density, has finite second moment, and is square integrable.
A3. $h = h_n$ is deterministic, $h \rightarrow 0$, and $nh \rightarrow \infty$.

2.5 Exact expectation and variance

By linearity and convolution,

$$\mathbb{E}[\hat{f}(x; h)] = \int K_h(x - y)f(y) dy = (K_h * f)(x),$$

$$\text{Var}\{\hat{f}(x; h)\} = \frac{1}{n} \{(K_h^2 * f)(x) - (K_h * f)(x)^2\}.$$

The estimator is biased for fixed h because $(K_h * f)(x) \neq f(x)$ in general.

2.6 Asymptotic bias, variance, and MSE

Using $z = (x - y)/h$,

$$\mathbb{E}[\hat{f}(x; h)] = \int K(z)f(x - hz) dz.$$

A second-order Taylor expansion gives

$$f(x - hz) = f(x) - hf'(x)z + \frac{1}{2}h^2 f''(x)z^2 + o(h^2 z^2).$$

Because K is symmetric,

$$\int zK(z) dz = 0.$$

Hence, under A1–A3,

$$\begin{aligned} \text{Bias}\{\hat{f}(x; h)\} &= \frac{1}{2}\mu_2(K)f''(x)h^2 + o(h^2), \\ \text{Var}\{\hat{f}(x; h)\} &= \frac{R(K)}{nh}f(x) + o((nh)^{-1}). \end{aligned}$$

Therefore,

$$\text{MSE}\{\hat{f}(x; h)\} = \frac{\mu_2(K)^2}{4}f''(x)^2h^4 + \frac{R(K)}{nh}f(x) + o(h^4 + (nh)^{-1}).$$

Equivalently,

$$\hat{f}(x; h) = f(x) + O(h^2) + O_{\mathbb{P}}((nh)^{-1/2}).$$

Thus $\hat{f}(x; h) \xrightarrow{\mathbb{P}} f(x)$ under A1–A3.

Here are some interpretations of the bias.

- If $f''(x) < 0$, then f is locally concave and the leading bias is negative. Peaks tend to be underestimated.
- If $f''(x) > 0$, then f is locally convex and the leading bias is positive. Valleys and tails tend to be overestimated.
- Larger curvature means a harder density estimation problem.

Also, we can interpret the variance as below.

$$\text{Var}\{\widehat{f}(x; h)\} \approx \frac{R(K)}{nh} f(x).$$

The quantity nh acts as an effective sample size.

2.7 Asymptotic normality

Assume additionally that

$$\int K(z)^{2+\delta} dz < \infty \quad \text{for some } \delta > 0.$$

Then

$$\sqrt{nh} \left\{ \widehat{f}(x; h) - \mathbb{E}[\widehat{f}(x; h)] \right\} \xrightarrow{d} N(0, R(K)f(x)).$$

If also $nh^5 = O(1)$, then

$$\sqrt{nh} \left\{ \widehat{f}(x; h) - f(x) - \frac{1}{2}\mu_2(K)f''(x)h^2 \right\} \xrightarrow{d} N(0, R(K)f(x)).$$

The rate is $\sqrt{nh}$, not $\sqrt{n}$.

2.8 Integrated error criteria

$$\text{ISE}\{\widehat{f}(\cdot; h)\} = \int \{\widehat{f}(x; h) - f(x)\}^2 dx,$$

$$\text{MISE}\{\widehat{f}(\cdot; h)\} = \mathbb{E}[\text{ISE}\{\widehat{f}(\cdot; h)\}] = \int \text{MSE}\{\widehat{f}(x; h)\} dx,$$

$$\text{MIAE}\{\widehat{f}(\cdot; h)\} = \mathbb{E} \int |\widehat{f}(x; h) - f(x)| dx.$$

The integrated absolute error is invariant under monotone transformations, but the squared error is easier to analyze.

Integrating the pointwise MSE gives

$$\text{MISE}\{\widehat{f}(\cdot; h)\} = \frac{1}{4}\mu_2(K)^2 R(f'')h^4 + \frac{R(K)}{nh} + o(h^4 + (nh)^{-1}).$$

The leading part is

$$\text{AMISE}\{\widehat{f}(\cdot; h)\} = \frac{1}{4}\mu_2(K)^2 R(f'')h^4 + \frac{R(K)}{nh}.$$

The minimizing bandwidth is

$$h_{\text{AMISE}} = \left\{ \frac{R(K)}{\mu_2(K)^2 R(f'')n} \right\}^{1/5}.$$

The optimal order is

$$\inf_{h>0} \text{AMISE}\{\widehat{f}(\cdot; h)\} = O(n^{-4/5}), \quad h_{\text{AMISE}} = O(n^{-1/5}).$$

2.9 Bandwidth selection

2.9.1 Normal scale and rule of thumb

Under the normal kernel density,

$$R\{\phi''_{\sigma}(\cdot - \mu)\} = \frac{3}{8\sqrt{\pi}\sigma^5}.$$

Thus

$$\hat{h}_{\text{NS}} = \left\{ \frac{8\sqrt{\pi}R(K)}{3\mu_2(K)^2n} \right\}^{1/5} \hat{\sigma}.$$

A robust scale estimate is

$$\hat{\sigma} = \min(s, \hat{\sigma}_{\text{IQR}}), \quad \hat{\sigma}_{\text{IQR}} = \frac{X_{[0.75n]} - X_{[0.25n]}}{\Phi^{-1}(0.75) - \Phi^{-1}(0.25)}.$$

For the normal kernel, $\mu_2(K) = 1$ and $R(K) = 1/(2\sqrt{\pi})$, so

$$\hat{h}_{\text{RT}} = \left(\frac{4}{3}\right)^{1/5} n^{-1/5} \hat{\sigma} \approx 1.06n^{-1/5} \hat{\sigma}.$$

This selector is simple but may oversmooth a density that is far from normal.

2.9.2 Direct plug-in selector

Repeated integration by parts gives

$$R(f^{(s)}) = (-1)^s \int f^{(2s)}(x) f(x) dx.$$

Define

$$\psi_r = \int f^{(r)}(x) f(x) dx = \mathbb{E}[f^{(r)}(X)].$$

Then $R(f'') = \psi_4$. A kernel estimator of ψ_r is

$$\hat{\psi}_r(g) = \frac{1}{n} \sum_{i=1}^n \hat{f}^{(r)}(X_i; g) = \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n L_g^{(r)}(X_i - X_j).$$

For a normal kernel and even r ,

$$\psi_r = \frac{(-1)^{r/2} r!}{(2\sigma)^{r+1} (r/2)! \sqrt{\pi}}.$$

The usual two-stage DPI construction is

$$\begin{aligned} \hat{\psi}_8^{\text{NS}} &= \frac{105}{32\sqrt{\pi}\hat{\sigma}^9}, \\ g_1 &= \left\{ -\frac{2K^{(6)}(0)}{\mu_2(K)\hat{\psi}_8^{\text{NS}}n} \right\}^{1/9}, \quad \hat{\psi}_6(g_1), \\ g_2 &= \left\{ -\frac{2K^{(4)}(0)}{\mu_2(K)\hat{\psi}_6(g_1)n} \right\}^{1/7}, \quad \hat{\psi}_4(g_2), \\ \hat{h}_{\text{DPI}} &= \left\{ \frac{R(K)}{\mu_2(K)^2\hat{\psi}_4(g_2)n} \right\}^{1/5}. \end{aligned}$$

2.9.3 Least-squares cross-validation

Expand the MISE as

$$\text{MISE} = \mathbb{E} \int \hat{f}^2 - 2\mathbb{E} \int \hat{f}f + \int f^2.$$

The last term does not depend on h . Define the leave-one-out estimator

$$\hat{f}_{-i}(x; h) = \frac{1}{n-1} \sum_{j \neq i} K_h(x - X_j).$$

The least-squares cross-validation criterion is

$$\text{LSCV}(h) = \int \hat{f}(x; h)^2 dx - \frac{2}{n} \sum_{i=1}^n \hat{f}_{-i}(X_i; h).$$

It satisfies

$$\mathbb{E}[\text{LSCV}(h)] = \text{MISE}\{\hat{f}(\cdot; h)\} - R(f).$$

Thus

$$\hat{h}_{\text{LSCV}} = \arg \min_{h>0} \text{LSCV}(h).$$

The objective may have several local minima, so a grid check is useful.

2.9.4 Biased cross-validation

A leave-out-diagonal estimate of $R(f'')$ is

$$\widehat{R(f'')} = R\{\hat{f}''(\cdot; h)\} - \frac{R(K'')}{nh^5} = \frac{1}{n^2} \sum_{i=1}^n \sum_{j \neq i} (K_h'' * K_h'')(X_i - X_j).$$

Then

$$\begin{aligned} \text{BCV}(h) &= \frac{1}{4} \mu_2(K)^2 \widehat{R(f'')} h^4 + \frac{R(K)}{nh}, \\ \hat{h}_{\text{BCV}} &= \arg \text{locmin}_{h>0} \text{BCV}(h), \end{aligned}$$

where the smallest local minimizer is used. BCV is usually less variable but more biased than LSCV.

2.9.5 Comparison

Under regularity conditions,

$$\begin{aligned} n^{1/10} \left(\frac{\hat{h}_{\text{LSCV}}}{h_{\text{MISE}}} - 1 \right) &\xrightarrow{d} N(0, \sigma_{\text{LSCV}}^2), \\ n^{1/10} \left(\frac{\hat{h}_{\text{BCV}}}{h_{\text{MISE}}} - 1 \right) &\xrightarrow{d} N(0, \sigma_{\text{BCV}}^2), \\ n^{5/14} \left(\frac{\hat{h}_{\text{DPI}}}{h_{\text{MISE}}} - 1 \right) &\xrightarrow{d} N(0, \sigma_{\text{DPI}}^2). \end{aligned}$$

Thus DPI converges faster. LSCV is flexible but variable. BCV is more stable and tends to choose a larger bandwidth.

2.10 Boundary correction by transformation

If the support has a boundary, an ordinary kernel spreads mass outside the support. Near a boundary, the bias is generally of order $O(h)$ rather than $O(h^2)$.

Let $Y = t(X)$ have density g . Then

$$f(x) = g(t(x))t'(x).$$

Estimate g on the transformed scale and return to the original scale:

$$\hat{f}_T(x; h, t) = \frac{1}{n} \sum_{i=1}^n K_h\{t(x) - t(X_i)\}t'(x).$$

Typical choices are the log transformation for (a, ∞) and the probit transformation for (a, b) .

2.11 Sampling from a KDE

Since

$$\hat{f}(x; h) = \frac{1}{n} \sum_{i=1}^n K_h(x - X_i),$$

sampling from $\hat{f}$ is sampling from a finite mixture:

1. choose I uniformly from $\{1, \dots, n\}$;
2. sample $Z \sim K$ independently;
3. set $X^* = X_I + hZ$.

3 Chapter 4: Univariate Kernel Regression

Let $(X_1, Y_1), \dots, (X_n, Y_n)$ be iid observations. The target is the regression function

$$m(x) = \mathbb{E}[Y \mid X = x].$$

If (X, Y) is continuous, then

$$m(x) = \int y f_{Y|X=x}(y) dy = \frac{\int y f(x, y) dy}{f_X(x)}.$$

3.1 Nadaraya–Watson estimator

Use a product kernel estimate of the joint density,

$$\hat{f}(x, y; h_1, h_2) = \frac{1}{n} \sum_{i=1}^n K_{h_1}(x - X_i) K_{h_2}(y - Y_i),$$

and

$$\hat{f}_X(x; h_1) = \frac{1}{n} \sum_{i=1}^n K_{h_1}(x - X_i).$$

Then

$$\frac{\int y \hat{f}(x, y; h_1, h_2) dy}{\hat{f}_X(x; h_1)} = \frac{\sum_{i=1}^n K_{h_1}(x - X_i) \int y K_{h_2}(y - Y_i) dy}{\sum_{j=1}^n K_{h_1}(x - X_j)}$$

$$= \frac{\sum_{i=1}^n K_{h_1}(x - X_i) Y_i}{\sum_{j=1}^n K_{h_1}(x - X_j)}.$$

The bandwidth in the Y direction disappears. Writing $h = h_1$,

$$\hat{m}(x; 0, h) = \sum_{i=1}^n W_i^0(x) Y_i, \quad W_i^0(x) = \frac{K_h(x - X_i)}{\sum_{j=1}^n K_h(x - X_j)}.$$

Since $W_i^0(x) \geq 0$ and $\sum_i W_i^0(x) = 1$, this is a local weighted mean. It is also called the local constant estimator.

3.2 Local polynomial estimator

For X_i near x ,

$$m(X_i) \approx m(x) + m'(x)(X_i - x) + \cdots + \frac{m^{(p)}(x)}{p!}(X_i - x)^p.$$

Replace the unknown derivatives by coefficients $\beta_0, \dots, \beta_p$ and solve

$$\hat{\beta}_h = \arg \min_{\beta \in \mathbb{R}^{p+1}} \sum_{i=1}^n \left\{ Y_i - \sum_{j=0}^p \beta_j (X_i - x)^j \right\}^2 K_h(x - X_i).$$

Define

$$\mathbf{X}_x = \begin{pmatrix} 1 & X_1 - x & \cdots & (X_1 - x)^p \\ \vdots & \vdots & & \vdots \\ 1 & X_n - x & \cdots & (X_n - x)^p \end{pmatrix},$$

$$\mathbf{W}_x = \text{diag}\{K_h(X_1 - x), \dots, K_h(X_n - x)\}, \quad \mathbf{Y} = (Y_1, \dots, Y_n)^\top.$$

Then

$$\hat{\beta}_h = (\mathbf{X}_x^\top \mathbf{W}_x \mathbf{X}_x)^{-1} \mathbf{X}_x^\top \mathbf{W}_x \mathbf{Y}.$$

The local polynomial estimate is the fitted intercept:

$$\hat{m}(x; p, h) = \hat{\beta}_{h,0} = \mathbf{e}_1^\top (\mathbf{X}_x^\top \mathbf{W}_x \mathbf{X}_x)^{-1} \mathbf{X}_x^\top \mathbf{W}_x \mathbf{Y} = \sum_{i=1}^n W_i^p(x) Y_i.$$

For $p = 0$, this reduces to Nadaraya–Watson. For $p = 1$, define

$$\hat{s}_r(x; h) = \frac{1}{n} \sum_{i=1}^n (X_i - x)^r K_h(x - X_i), \quad r = 0, 1, 2.$$

Then the local linear weights are

$$W_i^1(x) = \frac{1}{n} \frac{\hat{s}_2(x; h) - \hat{s}_1(x; h)(X_i - x)}{\hat{s}_2(x; h)\hat{s}_0(x; h) - \hat{s}_1(x; h)^2} K_h(x - X_i).$$

The weights sum to one, but they need not be nonnegative.

3.3 Regression model and assumptions

Write the location–scale model as

$$Y = m(X) + \sigma(X)\varepsilon, \quad \mathbb{E}[\varepsilon \mid X] = 0, \quad \text{Var}(\varepsilon \mid X) = 1,$$

so that

$$\sigma^2(x) = \text{Var}(Y \mid X = x).$$

Let f denote the marginal density of X .

A1. m is twice continuously differentiable.

A2. σ^2 is continuous and positive.

A3. f is continuously differentiable and bounded away from zero on the region of interest.

A4. K is a symmetric and bounded density, has finite second moment, and is square integrable.

A5. $h = h_n$ is deterministic, $h \rightarrow 0$, and $nh \rightarrow \infty$.

3.4 Conditional bias and variance

Condition on $X_1, \dots, X_n$. For $p = 0, 1$,

$$\text{Bias}\{\hat{m}(x; p, h) \mid X_1, \dots, X_n\} = B_p(x)h^2 + o_{\mathbb{P}}(h^2),$$

$$\text{Var}\{\hat{m}(x; p, h) \mid X_1, \dots, X_n\} = \frac{R(K)}{nhf(x)}\sigma^2(x) + o_{\mathbb{P}}((nh)^{-1}),$$

where

$$B_0(x) = \frac{\mu_2(K)}{2} \left\{ m''(x) + 2 \frac{m'(x)f'(x)}{f(x)} \right\},$$

$$B_1(x) = \frac{\mu_2(K)}{2} m''(x).$$

Local constant versus local linear.

- Both have bias of order h^2 and variance of order $(nh)^{-1}$.
- The local constant bias contains the extra design term $2m'(x)f'(x)/f(x)$.
- The local linear estimator removes this term and has the same first-order variance.
- A local constant fit is exactly unbiased for a constant regression function.
- A local linear fit is exactly unbiased for a linear regression function.
- More generally, an odd local polynomial order is usually preferred to the preceding even order.

The variance has the form

$$\text{Var}\{\hat{m}(x; p, h) \mid X_1, \dots, X_n\} \approx \frac{R(K)\sigma^2(x)}{nhf(x)}.$$

Thus low predictor density and large conditional variance both increase uncertainty.

3.5 Asymptotic normality

Assume that for some $\delta > 0$,

$$\mathbb{E}\{|Y - m(x)|^{2+\delta} \mid X = x\} < \infty.$$

Then, under A1–A5,

$$\sqrt{nh} \{\hat{m}(x; p, h) - \mathbb{E}[\hat{m}(x; p, h) \mid X_1, \dots, X_n]\} \xrightarrow{d} N\left(0, \frac{R(K)\sigma^2(x)}{f(x)}\right).$$

If also $nh^5 = O(1)$, then

$$\sqrt{nh} \{\hat{m}(x; p, h) - m(x) - B_p(x)h^2\} \xrightarrow{d} N\left(0, \frac{R(K)\sigma^2(x)}{f(x)}\right).$$

3.6 Bandwidth selection for local polynomial regression

A density-weighted integrated squared error is

$$\text{ISE}\{\hat{m}(\cdot; p, h)\} = \int \{\hat{m}(x; p, h) - m(x)\}^2 f(x) dx.$$

Conditioning on the predictors,

$$\text{MISE}\{\hat{m}(\cdot; p, h) \mid X_1, \dots, X_n\} = \int \text{MSE}\{\hat{m}(x; p, h) \mid X_1, \dots, X_n\} f(x) dx.$$

The leading asymptotic error is

$$\text{AMISE}\{\hat{m}(\cdot; p, h) \mid X_1, \dots, X_n\} = h^4 \int B_p(x)^2 f(x) dx + \frac{R(K)}{nh} \int \sigma^2(x) dx.$$

For the local linear estimator,

$$\theta_{22}(m) = \int m''(x)^2 f(x) dx,$$

so

$$h_{\text{AMISE}} = \left\{ \frac{R(K) \int \sigma^2(x) dx}{\mu_2(K)^2 \theta_{22}(m) n} \right\}^{1/5}.$$

3.6.1 Rule-of-thumb selector

Fit a global quartic model

$$\hat{m}_Q(x) = \hat{\alpha}_0 + \hat{\alpha}_1 x + \hat{\alpha}_2 x^2 + \hat{\alpha}_3 x^3 + \hat{\alpha}_4 x^4.$$

Then

$$\hat{m}_Q''(x) = 2\hat{\alpha}_2 + 6\hat{\alpha}_3 x + 12\hat{\alpha}_4 x^2,$$

$$\hat{\theta}_{22}(\hat{m}_Q) = \frac{1}{n} \sum_{i=1}^n \hat{m}_Q''(X_i)^2.$$

Under homoscedasticity,

$$\hat{\sigma}_Q^2 = \frac{1}{n-5} \sum_{i=1}^n \{Y_i - \hat{m}_Q(X_i)\}^2,$$

and approximate

$$\int \sigma^2(x) dx \approx \{X_{(n)} - X_{(1)}\} \hat{\sigma}_Q^2.$$

This gives

$$\hat{h}_{\text{RT}} = \left\{ \frac{R(K) \{X_{(n)} - X_{(1)}\} \hat{\sigma}_Q^2}{\mu_2(K)^2 \hat{\theta}_{22}(\hat{m}_Q) n} \right\}^{1/5}.$$

Its quality depends on the quartic approximation and the homoscedasticity assumption.

3.6.2 Cross-validation

The ordinary residual criterion

$$\frac{1}{n} \sum_{i=1}^n \{Y_i - \hat{m}(X_i; p, h)\}^2$$

favors $h \approx 0$ because a very small bandwidth can nearly interpolate the data.

Instead, define the leave-one-out estimator $\hat{m}_{-i}$ and

$$\text{CV}(h) = \frac{1}{n} \sum_{i=1}^n \{Y_i - \hat{m}_{-i}(X_i; p, h)\}^2, \quad \hat{h}_{\text{CV}} = \arg \min_{h>0} \text{CV}(h).$$

Because $\hat{m}$ is linear in the responses,

$$W_{-i,j}^p(x) = \frac{W_j^p(x)}{1 - W_i^p(x)}, \quad j \neq i,$$

and

$$\text{CV}(h) = \frac{1}{n} \sum_{i=1}^n \left\{ \frac{Y_i - \hat{m}(X_i; p, h)}{1 - W_i^p(X_i)} \right\}^2.$$

The CV objective may have local minima, so several starting values or a grid search are useful.

3.7 Naive regression estimator and regressogram

With the rectangular kernel,

$$\hat{m}_N(x; h) = \frac{\sum_{i=1}^n \mathbf{1}_{\{|X_i - x| < h\}} Y_i}{\sum_{i=1}^n \mathbf{1}_{\{|X_i - x| < h\}}} = \frac{1}{|\mathcal{N}(x; h)|} \sum_{i \in \mathcal{N}(x; h)} Y_i,$$

where

$$\mathcal{N}(x; h) = \{i : |X_i - x| < h\}.$$

It is undefined when $|\mathcal{N}(x; h)| = 0$ and is generally discontinuous.

For fixed bins $B_k = [t_k, t_{k+1})$,

$$\hat{m}_R(x; h) = \frac{\sum_{i=1}^n \mathbf{1}_{\{X_i \in B_k\}} Y_i}{\sum_{i=1}^n \mathbf{1}_{\{X_i \in B_k\}}} = \frac{1}{|\mathcal{B}(k; h)|} \sum_{i \in \mathcal{B}(k; h)} Y_i, \quad x \in B_k.$$

This is the regressogram. It is constant on each fixed bin and is more rigid than the moving-window estimator.

3.8 Behavior in low-density regions

Both local constant and local linear estimators can be unstable where $f(x)$ is small or where the support of X has a gap. The local constant fit tends to stay near the response attached to the closest observation. The local linear fit extrapolates a local line and may produce large spikes. Nonparametric regression should therefore be interpreted mainly on regions supported by the predictor data.

4 Appendix A: Pointwise Confidence Intervals for a Density

4.1 Why a direct interval for $f(x)$ is difficult

For fixed h ,

$$\mathbb{E}[\hat{f}(x; h)] = (K_h * f)(x),$$

not $f(x)$. The difference

$$(K_h * f)(x) - f(x)$$

is the smoothing bias. Removing its leading term requires estimating $f''(x)$, which is harder than estimating $f(x)$ itself.

Therefore, the simple interval below targets the smoothed density

$$\mathbb{E}[\hat{f}(x; h)] = (K_h * f)(x),$$

not the original density $f(x)$.

4.2 Normal convolution example

For normal densities,

$$\{\phi_{\sigma_1}(\cdot - \mu_1) * \phi_{\sigma_2}(\cdot - \mu_2)\}(x) = \phi_{\sqrt{\sigma_1^2 + \sigma_2^2}}(x - \mu_1 - \mu_2).$$

Also,

$$\int \phi_{\sigma_1}(x - \mu_1) \phi_{\sigma_2}(x - \mu_2) dx = \phi_{\sqrt{\sigma_1^2 + \sigma_2^2}}(\mu_1 - \mu_2).$$

If $K = \phi$ and $f(x) = \phi_{\sigma}(x - \mu)$, then

$$\mathbb{E}[\hat{f}(x; h)] = (K_h * f)(x) = \phi_{\sqrt{h^2 + \sigma^2}}(x - \mu).$$

Thus the mean of the KDE is the density of $N(\mu, \sigma^2 + h^2)$.

The exact variance is

$$\text{Var}\{\hat{f}(x; h)\} = \frac{1}{n} \left\{ \frac{1}{2\sqrt{\pi}h} \phi_{\sqrt{h^2/2 + \sigma^2}}(x - \mu) - \phi_{\sqrt{h^2 + \sigma^2}}(x - \mu)^2 \right\}.$$

4.3 Asymptotic pointwise interval

From the KDE central limit theorem,

$$\sqrt{nh}\{\hat{f}(x; h) - \mathbb{E}[\hat{f}(x; h)]\} \xrightarrow{d} N(0, R(K)f(x)).$$

Since

$$\widehat{f}(x; h) = f(x)\{1 + o_{\mathbb{P}}(1)\},$$

Slutsky's theorem gives

$$\sqrt{\frac{nh}{R(K)\widehat{f}(x; h)}}\{\widehat{f}(x; h) - \mathbb{E}[\widehat{f}(x; h)]\} \xrightarrow{d} N(0, 1).$$

Hence an asymptotic pointwise $100(1 - \alpha)\%$ interval for the smoothed density is

$$I_{\alpha}(x) = \left[\widehat{f}(x; h) \pm z_{\alpha/2} \sqrt{\frac{R(K)\widehat{f}(x; h)}{nh}} \right].$$

Important remarks.

1. The target is $(K_h * f)(x)$, not $f(x)$.
2. The interval is pointwise:

$$\mathbb{P}\{(K_h * f)(x) \in I_{\alpha}(x)\} \approx 1 - \alpha$$

for a fixed x . This does not give simultaneous coverage over all x .

3. Good coverage needs both small bias and enough local data: h should be small and nh should be large.
4. The derivation treats h as deterministic. A data-driven bandwidth can change finite-sample coverage.
5. Replacing $f(x)$ by $\widehat{f}(x; h)$ is least reliable in very low-density regions.

For the normal kernel case, one may benchmark the interval above with the interval based on the exact variance,

$$\widehat{f}(x; h) \pm z_{\alpha/2} \sqrt{\text{Var}\{\widehat{f}(x; h)\}}.$$